

ZERUI GUO

✉ cyrus.guo@wisc.edu cyrus-guo.com 📍 Madison, WI, USA github.com/AlphaCat00

EDUCATION

University of Wisconsin-Madison, USA *May. 2023 - Present*
Ph.D. student in Computer Science, Advisor: Ming Liu

Beihang University (BUAA), China *Sep. 2020 - Jan. 2023*
Master of Computer Science and Technology, Advisor: Yuebin Bai

Beijing University of Post and Telecommunications (BUPT), China *Sep. 2016 - Jun. 2020*
Bachelor of Network Engineering, Ranking: 4/135
School Outstanding Graduate

RESEARCH INTERESTS

Memory Fabrics(CXL.mem), SmartNIC-assisted Computing

PUBLICATIONS

Building A CSFQ-Inspired Transport for Switched CXL Memory Pooling *NSDI 2026*
Zerui Guo, Emily Shriver, and Ming Liu

Understanding and Profiling CXL.mem Using PathFinder *SIGCOMM 2025*
Xiao Li, Zerui Guo, Yuebin Bai, Mahesh Ketkar, Hugh Wilkinson, and Ming Liu

Building Massive MIMO Baseband Processing on a Single-Node Supercomputer *NSDI 2025*
Xincheng Xie, Wentao Hou, Zerui Guo, and Ming Liu

LogNIC: A High-Level Performance Model for SmartNICs *MICRO 2023*
Zerui Guo, Jiaxin Lin, Yuebin Bai, Daehyeok Kim, Michael Swift, Aditya Akella, and Ming Liu

LEED: A Low-Power, Fast Persistent Key-Value Store on SmartNIC JBOFs *SIGCOMM 2023*
Zerui Guo, Hua Zhang, Chenxingyu Zhao, Yuebin Bai, Michael Swift, and Ming Liu

A Millimeter-wave Multi-channel MAC with Dynamic Spectrum Access Capability for Mobile Self-organizing Heterogeneous Networks *IPCCC 2021*
Peng Feng, Yuebin Bai, Zerui Guo, Xiaolin Wang, Yining Zhang, Bojian Bai, and Junmin Zhang

RESEARCH EXPERIENCE

A High-Level Performance Model for Programmable Network Hardware *Feb. 2022 - Apr. 2023*
University of Wisconsin-Madison

- Introduced a packet-centric modeling approach instead of an execution flow-based one to analyze the performance characteristics of SmartNIC-offloaded programs.
- Built a comprehensive model that abstracts low-level SmartNIC device details using a hardware model and represents offloaded programs as software execution graphs for estimating both throughput and latency.
- Validated the model's capabilities, including performance limit estimations, software optimizations, and guidance for hardware design, through extensive evaluation with commodity SmartNICs and an academic prototype across diverse application scenarios.

A Low-Power, Fast Persistent Key-Value Store on SmartNIC JBOFs

Sep. 2021 - Feb. 2023

University of Wisconsin-Madison

- Designed and implemented a circular log data structure and a DRAM/Flash hybrid indexing scheme to adapt to the highly-skewed storage hierarchy.
- Developed a token-based end-to-end I/O scheduler to prevent oversubscription of our system components by making early scheduling decisions.
- Proposed a data swapping mechanism and enhanced chain replication with request shipping capabilities to alleviate load imbalances at different levels.
- Achieved an average energy efficiency $4.2\times$ and $17.5\times$ higher than existing solutions that use beefy server JBOFs and wimpy embedded storage nodes in terms of requests per Joule.

TEACHING EXPERIENCE

Teaching Assistant, Embedded Systems, BUAA

2020 Fall, 2021 Spring

Designed and conducted in-class experiments from programming the firmware to writing custom driver module in Linux.

Teaching Assistant, Wireless Network Systems, BUAA

2020 Fall

Assisted with the experiment where students make custom modifications to the AODV-uu routing protocol to increase its anti-interference ability with machine learning.

Teaching Assistant, Introduction of Intelligent Car, BUPT

2019 Fall

Taught around 30 students weekly about various algorithms and techniques to build an intelligent trace-tracking car. Designed assignment sheets and graded them based on the cars they build.

WORK EXPERIENCE

University of Wisconsin-Madison, USA

Jun. 2023 - Now

Research Assistant

HONORS & AWARDS

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|---|------------------|
| - Second Prize, Beihang University Scholarship | 2021 |
| - Honorable Mention, Interdisciplinary Contest In Modeling | 2019 |
| - Second Prize, China "Internet plus" College Students' Innovation and Entrepreneurship Contest | 2019 |
| - Third Prize, The NXP Cup National University Students Intelligent Car Race | 2018 |
| - Second Prize, Beijing University of Post and Telecommunications Scholarship | 2016, 2017, 2018 |

SKILLS

Programming:	C/C++, Python, Golang, Java, JavaScript, Julia
Frameworks & Tools:	SPDK, RDMA, UCX, NVENC, Linux Networking, KVM, Docker, Django